

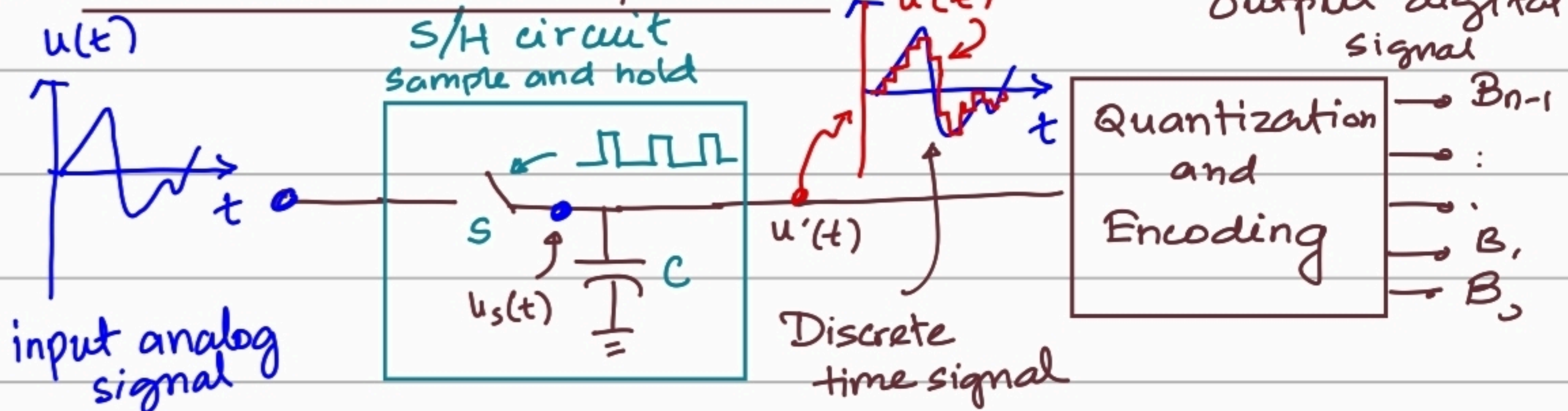
Lecture 16: ADC and DAC.

Physical world's informations are analog. Analog means data could take any real value at each instant of time.

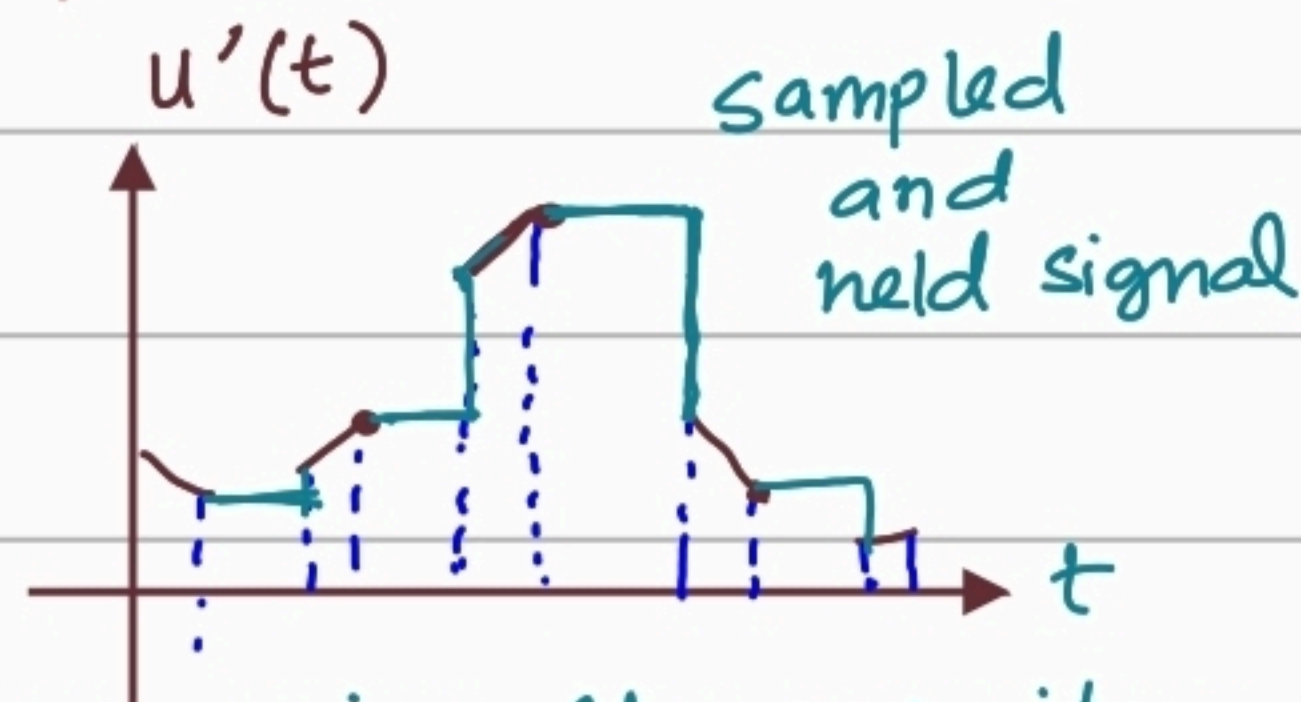
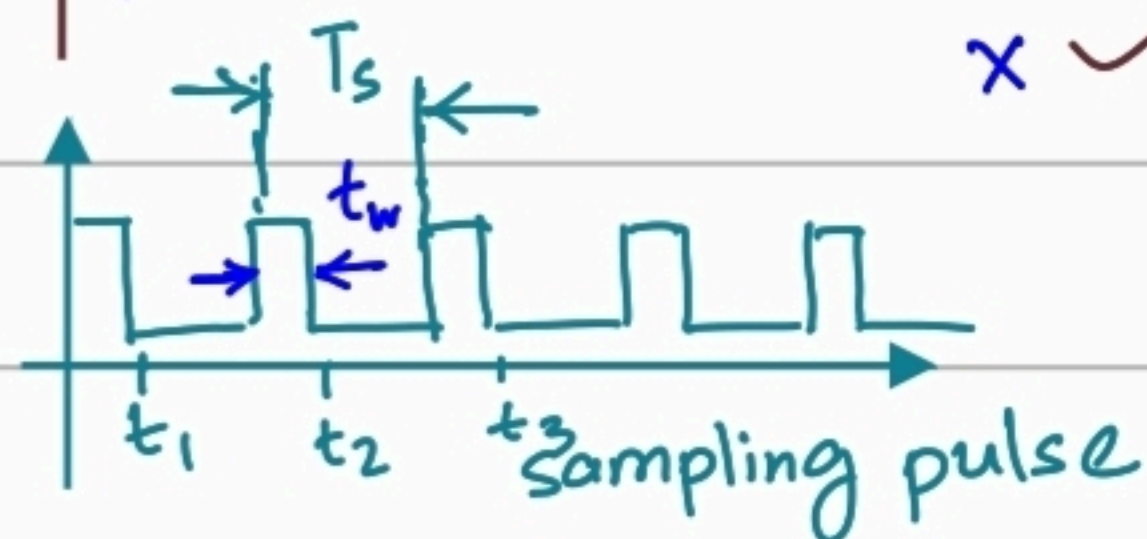
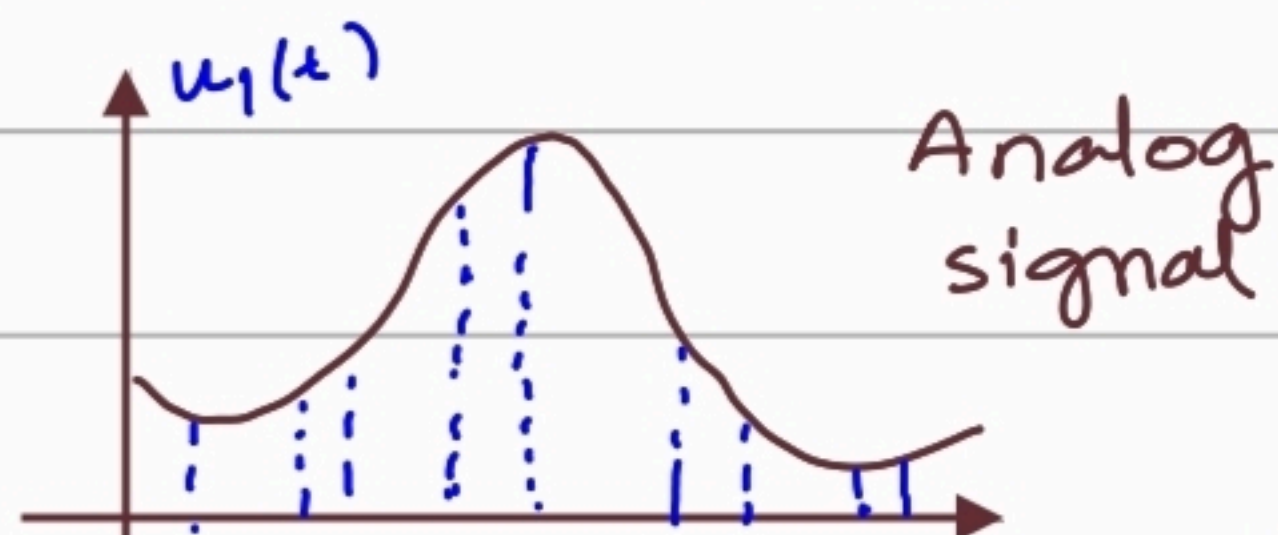
In order to store/process we use digital data in our computers and microprocessor. Digital data means binary bit of strings.

Environment $\rightarrow$ Transducer $\xrightarrow{\text{A/D}}$ Computer $\xrightarrow{\text{D/A}}$ speaker

A/D Conversion process:



Sample and Hold:

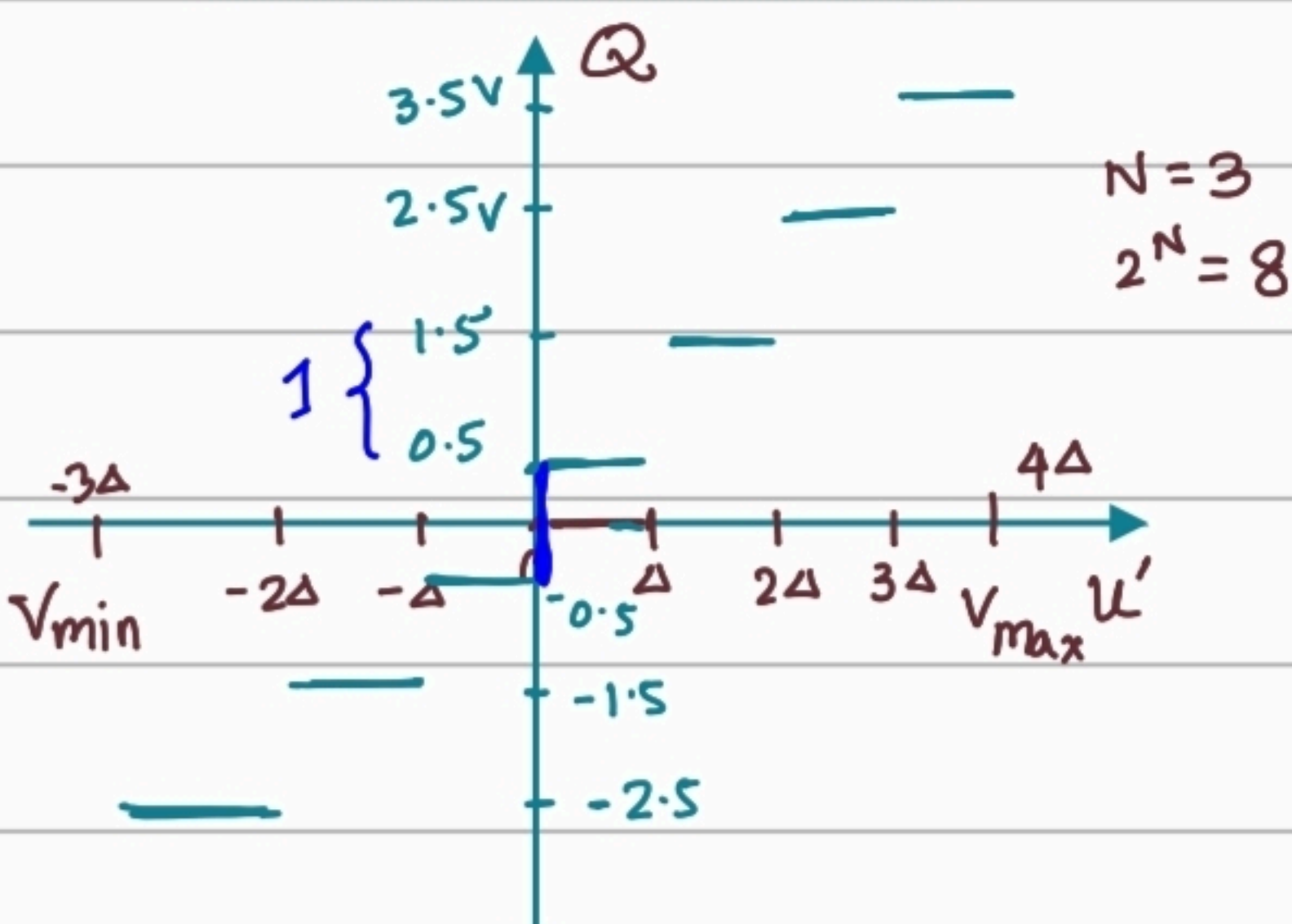


This hold operation is done using the capacitors.

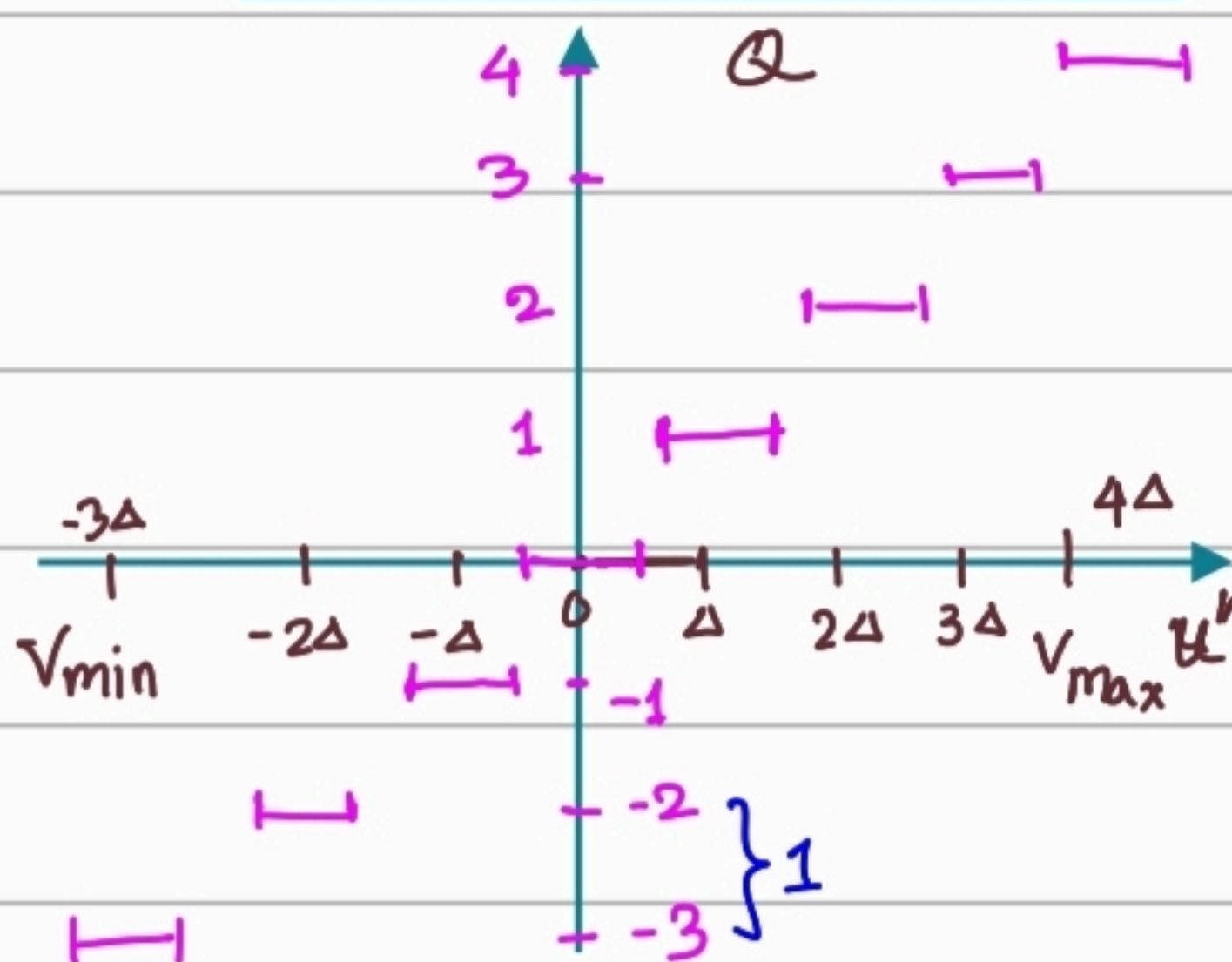
Quantization:

Quantization is process by which we assign sampled and hold signal data to some fixed preassigned values.

Midrise quantization



Mid tread quantization



N = number bits we want as digital output

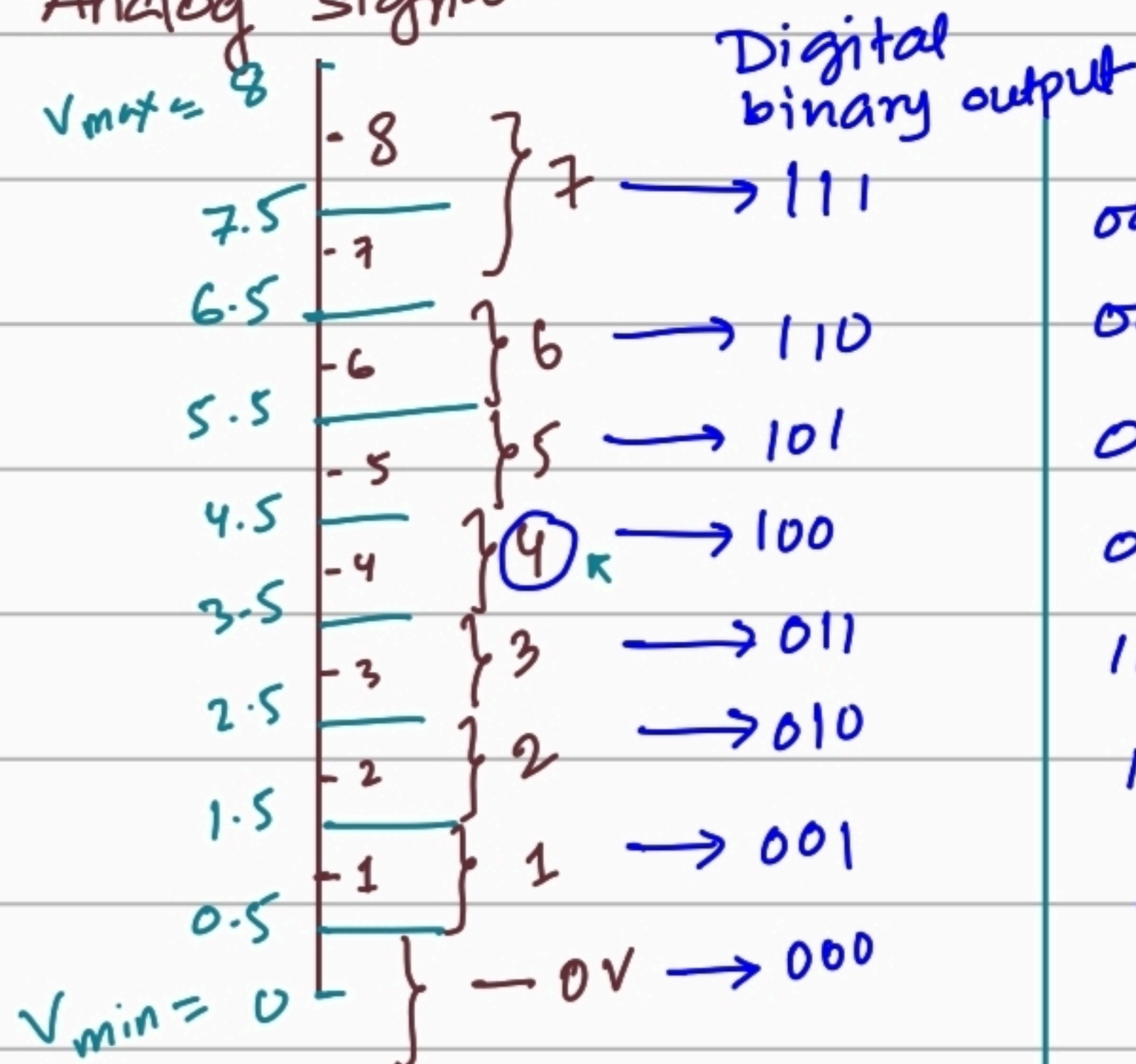
$$\Delta = \text{resolution} = \frac{V_{\max} - V_{\min}}{2^N}$$

2^N = number of levels.

$V_{\max} / V_{\min}$ = maximum/minimum possible value of analog signal.

Encoding: We can choose binary values for each quantized levels.

Analog signal.



$$\frac{V_{\max} + V_{\min}}{2} = 4$$

4 v is a quantization level so it is actually midtread quantization.

$$N=3, 2^N=8$$

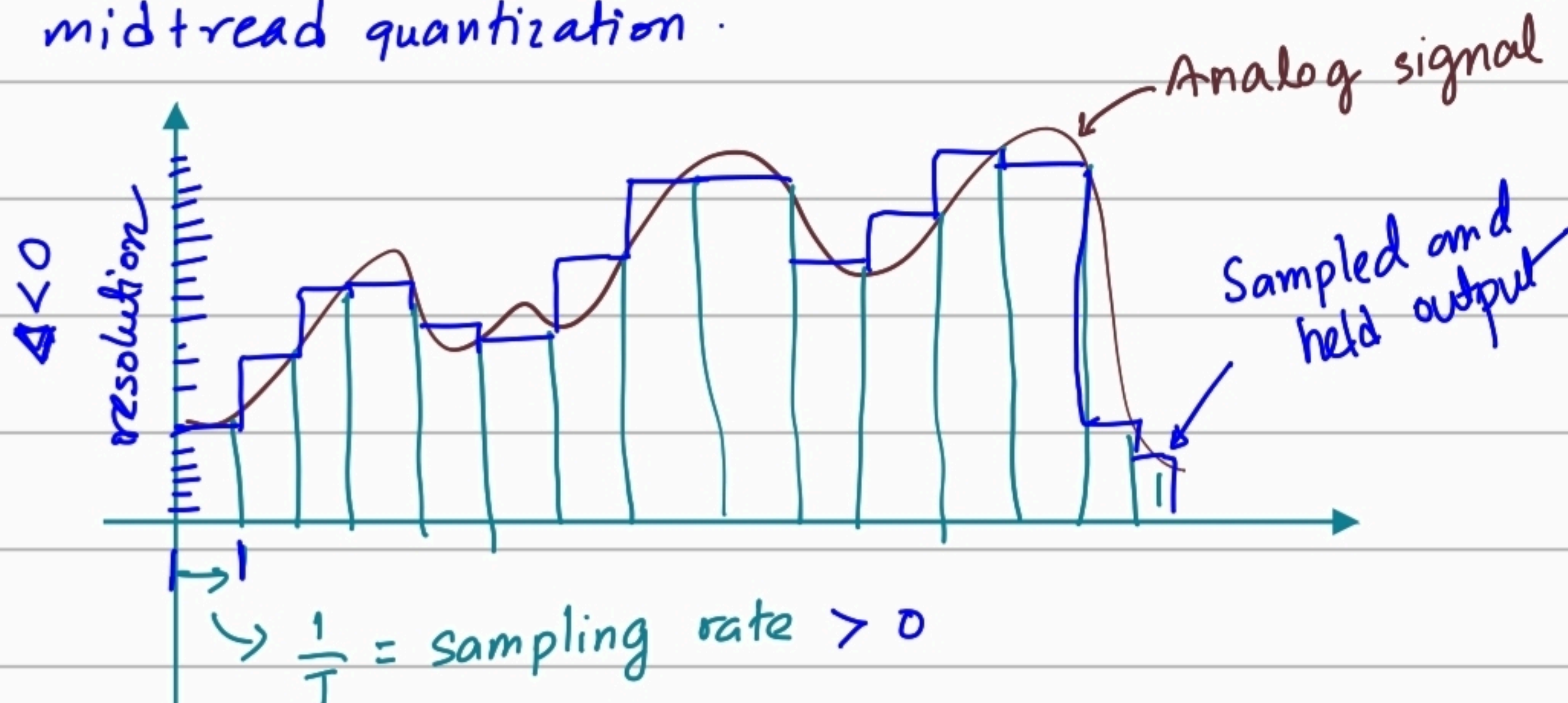
Midrise quantization

$$V_{\max} = 10V, V_{\min} = 0V$$

000	0 - 1.25V	→	0.625
001	1.25 - 2.5	→	1.875
010	2.5 - 3.75	→	3.125
011	3.75 - 5.0	→	4.375
100	5.0 - 6.25	→	5.625
101	6.25 - 7.5	→	6.875
110	7.5 - 8.75	→	8.125
111	8.75 - 10.0	→	9.375

$$\frac{V_{\min} + V_{\max}}{2} = 5V$$

Voltage levels



Flash A/D Converter:

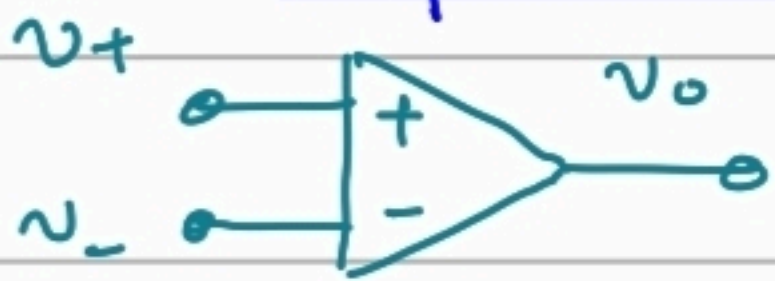
Adv. # This converter is very fast. It requires one clock cycle to convert the analog to digital data.

Dis. # It requires a lot component.

$$N = 3. 2^N = 8$$

Midrise quantization:

comparators



$v_+ > v_-$, $v_o = \text{High}$ ✓

$v_+ < v_-$, $v_o = \text{Low}$ ✓

$$V_{ref}^+ = 9.375 \text{ V}$$

$$V_{ref}^- = -0.675 \text{ V}$$

$$I = \frac{(V_{ref}^+ - V_{ref}^-)}{8R} = \frac{10}{8R}$$

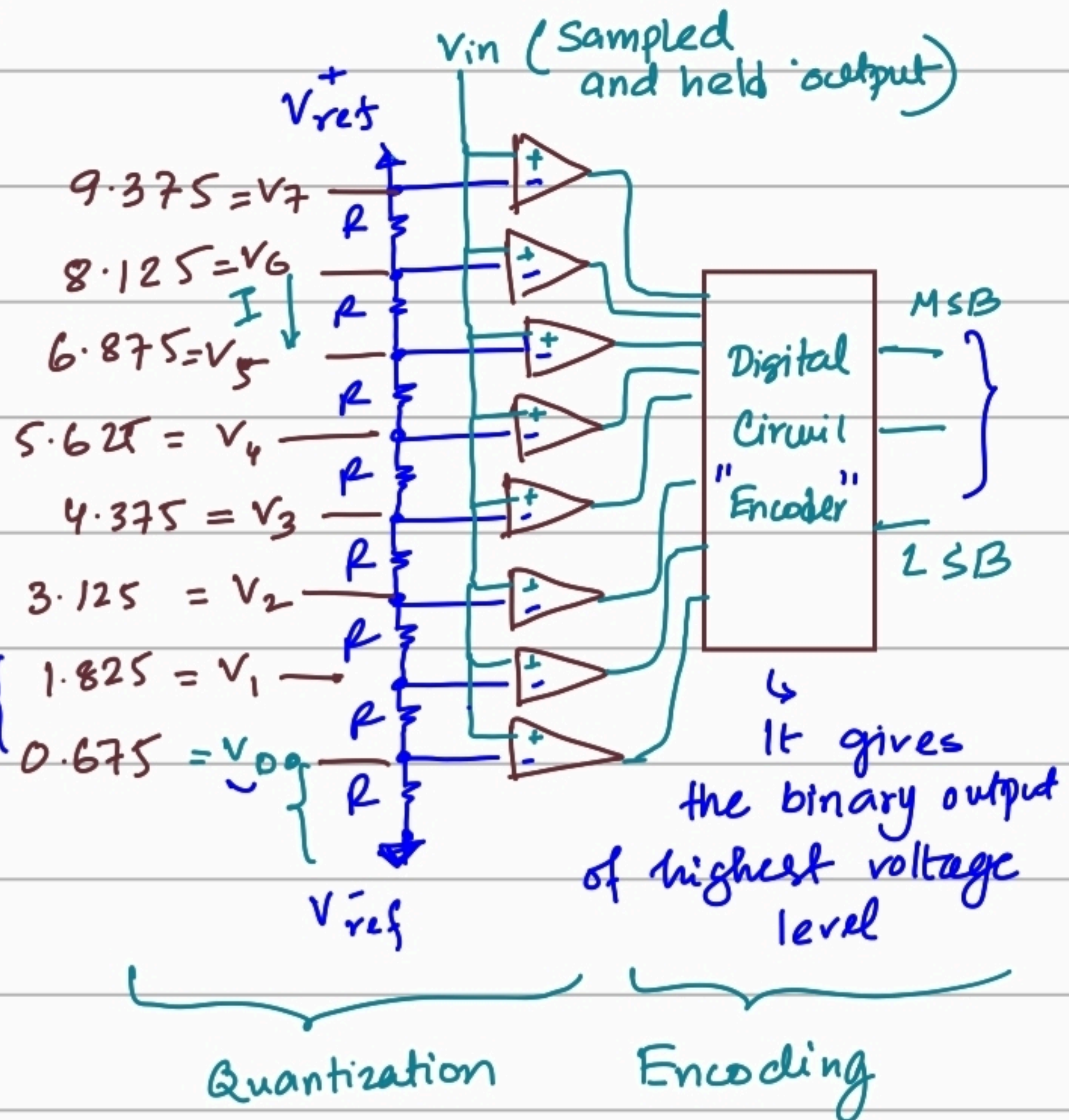
$$IR = v_o - V_{ref}^-$$

$$\Rightarrow v_o = V_{ref}^- + IR$$

$$= -0.675 + \frac{10}{8}$$

$$= 0.675$$

$$V_1 = v_o + IR = 1.825 \text{ V}$$



Dual Slope converter:

Adv. It requires small amount of circuit components. We can use many levels for quantization.

Disadvantage: Relatively slow.